MINNESOTA RECYCLING PATTERNS AND FINANCIAL TRENDS

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MIS 380, SECTION 1: BUSINESS INTELLIGENCE & ANALYTICS
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AI COLLABORATION & HUMAN-IN-THE-LOOP

Step 1: Baseline Research (ChatGpt)

Step 2: Document Grounding

Step 3: Synthesis & Narrative

Human-in-the-Loop: Final Verification

All AI-generated content was cross-referenced against the primary course resources (*Waste Management by Method, Expenditure and Revenue, and Recycling by Category per Professor Simon Jin*) by the authors to eliminate hallucinations and ensure technical validity

PROJECT OBJECTIVE

- Analyze recycling patterns across Minnesota counties
- Use MPCA datasets to evaluate waste management and recycling trends
- Identify relationships between recycling, expenditures, and policy changes
- Compare Minnesota recycling trends to national patterns

MINNESOTA POLLUTION CONTROL AGENCY

DATA OVERVIEW

1. **Waste Management by Method (1991-2017)**

- Recycling, organics, onsite disposal, waste to energy (WTE), landfilling (tons)
- County-level waste processing methods (burned or buried)

2. **Recycling by Category (1991-2017)**

- Materials: paper, glass, metal, plastic, hazardous waste, use oil
- Residential vs. Commercial/industrial (CII) recycling
- Total recycled tons per county

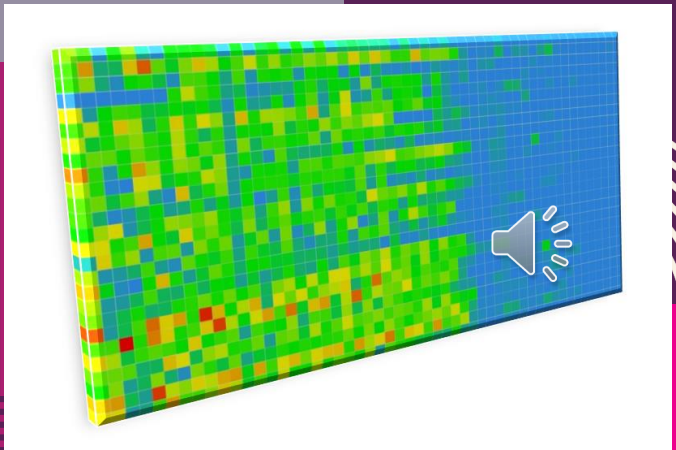
3. Expenditure & Revenue (2014-2017)

- Expenditures: administration, education, organics, recycling, waste-to-energy, etc.
- Revenue: SCORE funds, local revenue, other grants/material sales

Year	County	Expenditure Administration	Expenditure Education	Expenditure HHW Problem Material Mgmt	Expenditure Organics
2014	Aitkin	161,989	6,073	13,552	300
2014	Becker	295,359	16,133	133	21,863
2014	Beltrami	0	3,348	23,251	0
2014	Benton	232,003	22,008	34,583	0
2014	Big Stone	73,411	1,525	7,133	0
2014	Blue Earth	82,155	22,310	87,587	0
2014	Brown	56,805	2,125	56,302	1,125
2014	Carlton (Pa	79,837	4,716	43,535	3,108

DATA PREPARATION AND SAP PROCESS

- Import datasets into SAP Predictive Analytics
- Review variables and standardized data types
- Cleaned key fields (county names, year values)
- Calculated fields (total expenditure, total revenue, net income)
- Developed visualizations (heat maps, lines charts) to identify patterns



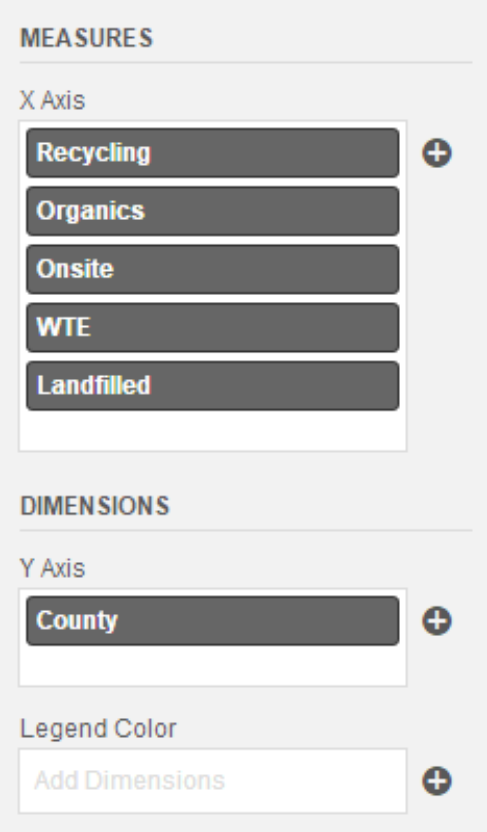
COUNTY PATTERNS: WASTE MANAGEMENT METHOD

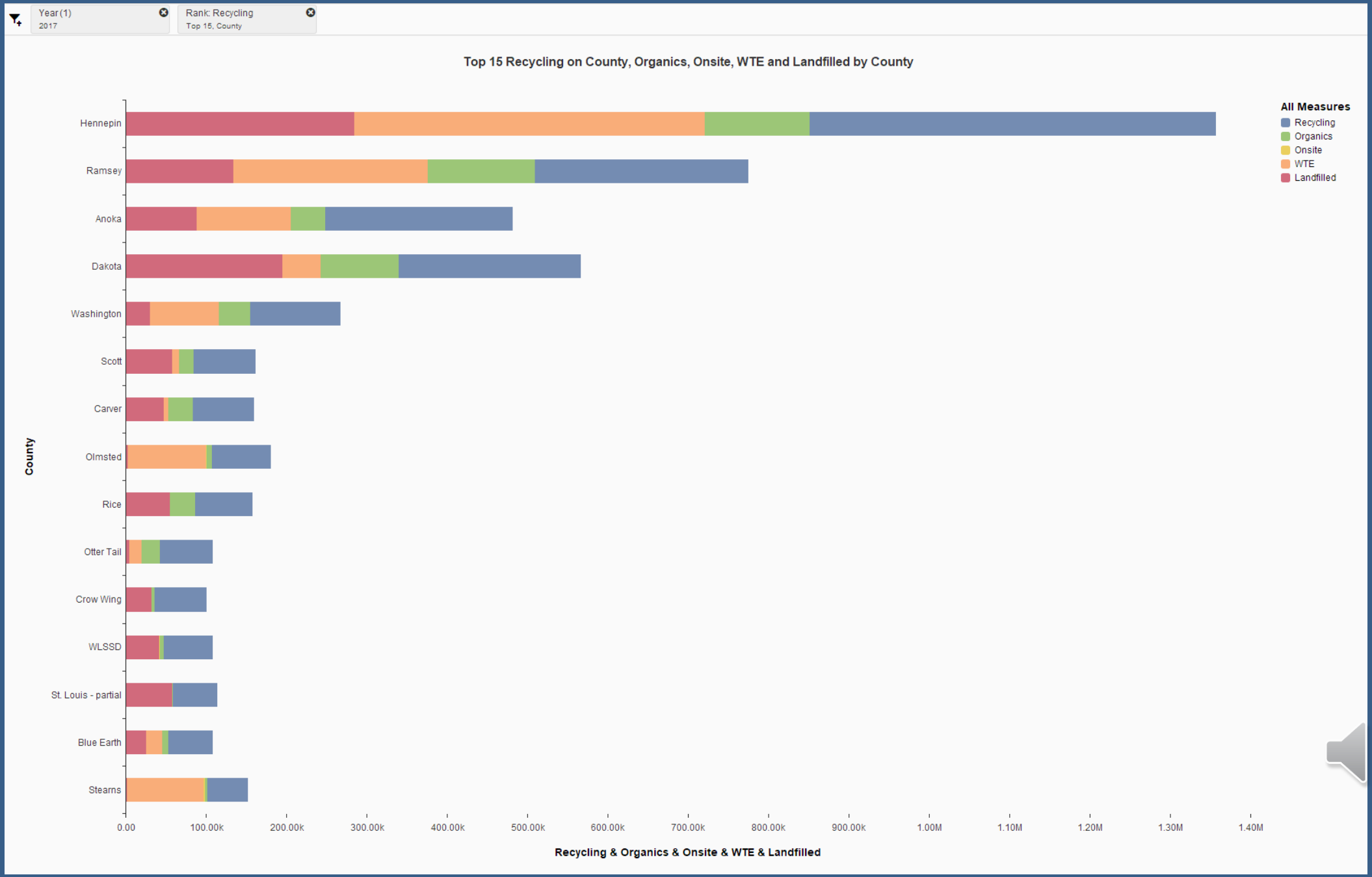
Question 1:

Are there different recycling patterns between counties for organics, WTE, landfilled, and other waste methods?

How:

- Stacked bar chart
- Measures & dimensions




Year (1)
2017
Rank: Recycling
Top 15, County
Top 15 Recycling on County, Organics, Onsite, WTE and Landfilled by County
All Measures
Recycling
Organics
Onsite
WTE
Landfilled
County
Hennepin
Ramsey
Anoka
Dakota
Washington
Scott
Carver
Olmsted
Rice
Otter Tail
Crow Wing
WLSSD
St. Louis - partial
Blue Earth
Stearns
0.00
100.00k
200.00k
300.00k
400.00k
500.00k
600.00k
700.00k
800.00k
900.00k
1.00M
1.10M
1.20M
1.30M
1.40M
Recycling & Organics & Onsite & WTE & Landfilled

COUNTY PATTERNS: WASTE MANAGEMENT METHOD

Answer:

Yes and no.

1. **Why:**
2. Only partially consistent
3. Varys from county to county

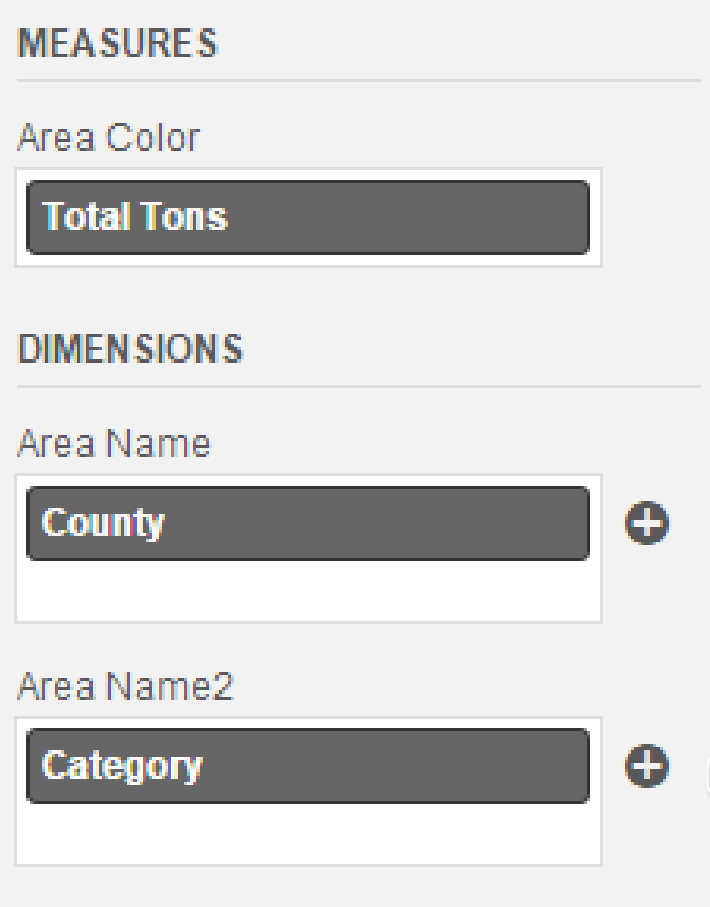
COUNTY PATTERNS: RECYCLING CATEGORY

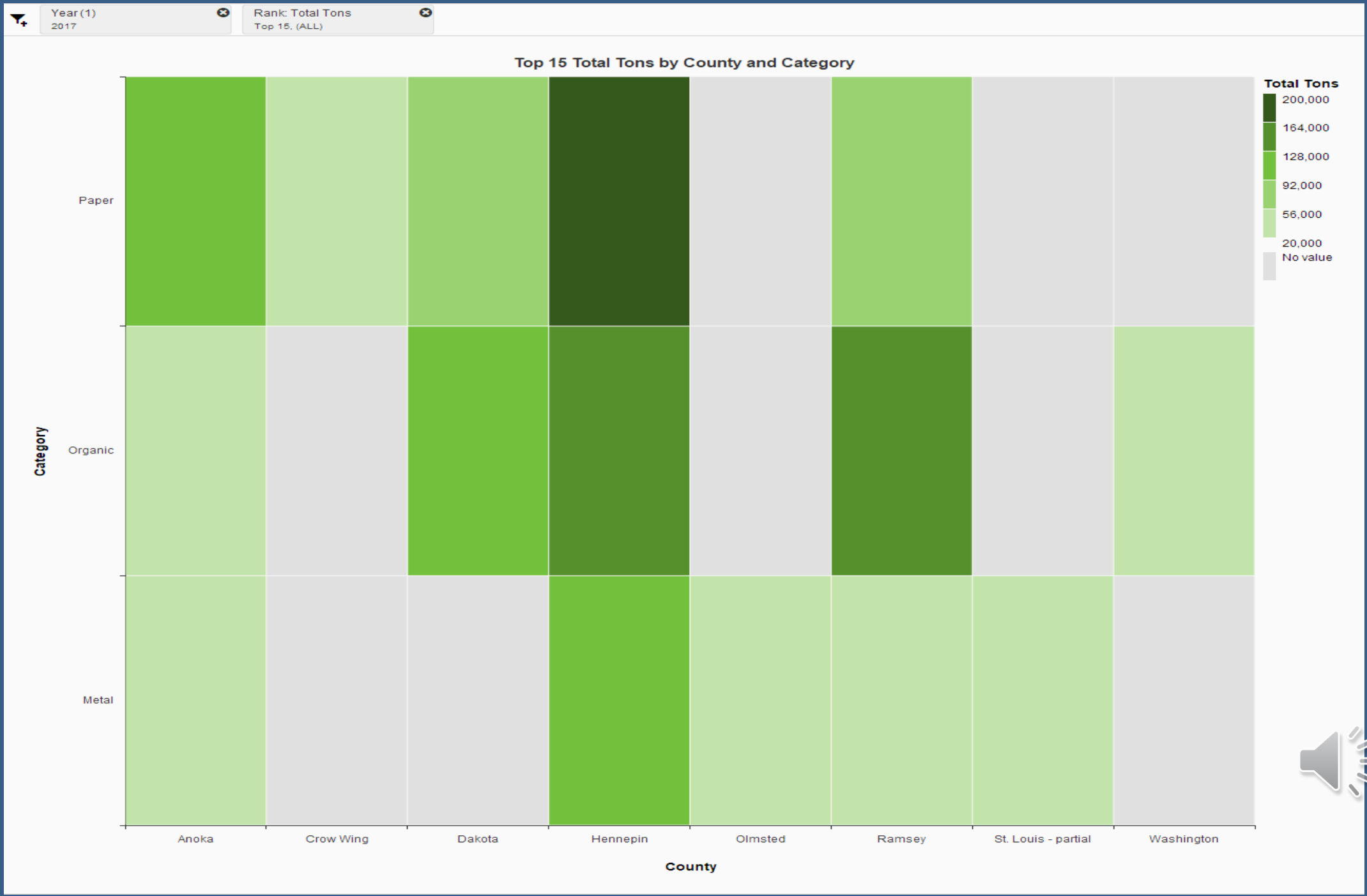
Question 2:

Are there different patterns between counties for paper, glass, metal, plastic, and other recycling categories?

How:

- Heat map
- Dimensions




Year (1)
2017
Rank: Total Tons
Top 15, (ALL)
Top 15 Total Tons by County and Category
Total Tons
200,000
164,000
128,000
92,000
56,000
20,000
No value
Category
Paper
Organic
Metal
Anoka
Crow Wing
Dakota
Hennepin
Olmsted
Ramsey
St. Louis - partial
Washington
County

COUNTY PATTERNS: RECYCLING CATEGORY

Answer:

Yes.

1. **Why:**
2. Only partially consistent
3. Varys from county to county
4. Inconsistency in category

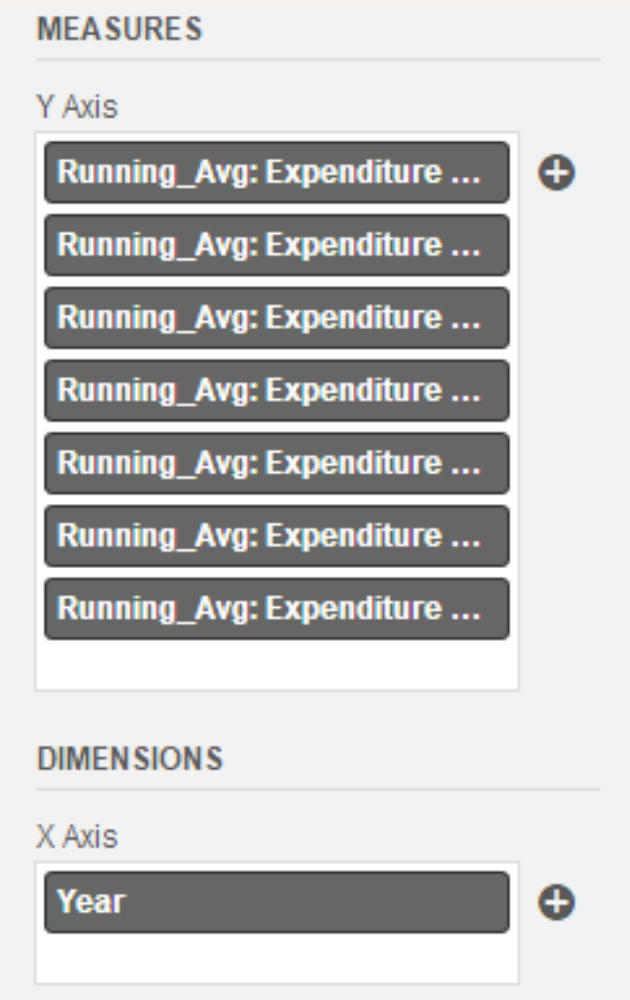
AVERAGE EXPENDITURE: CATEGORY & YEAR

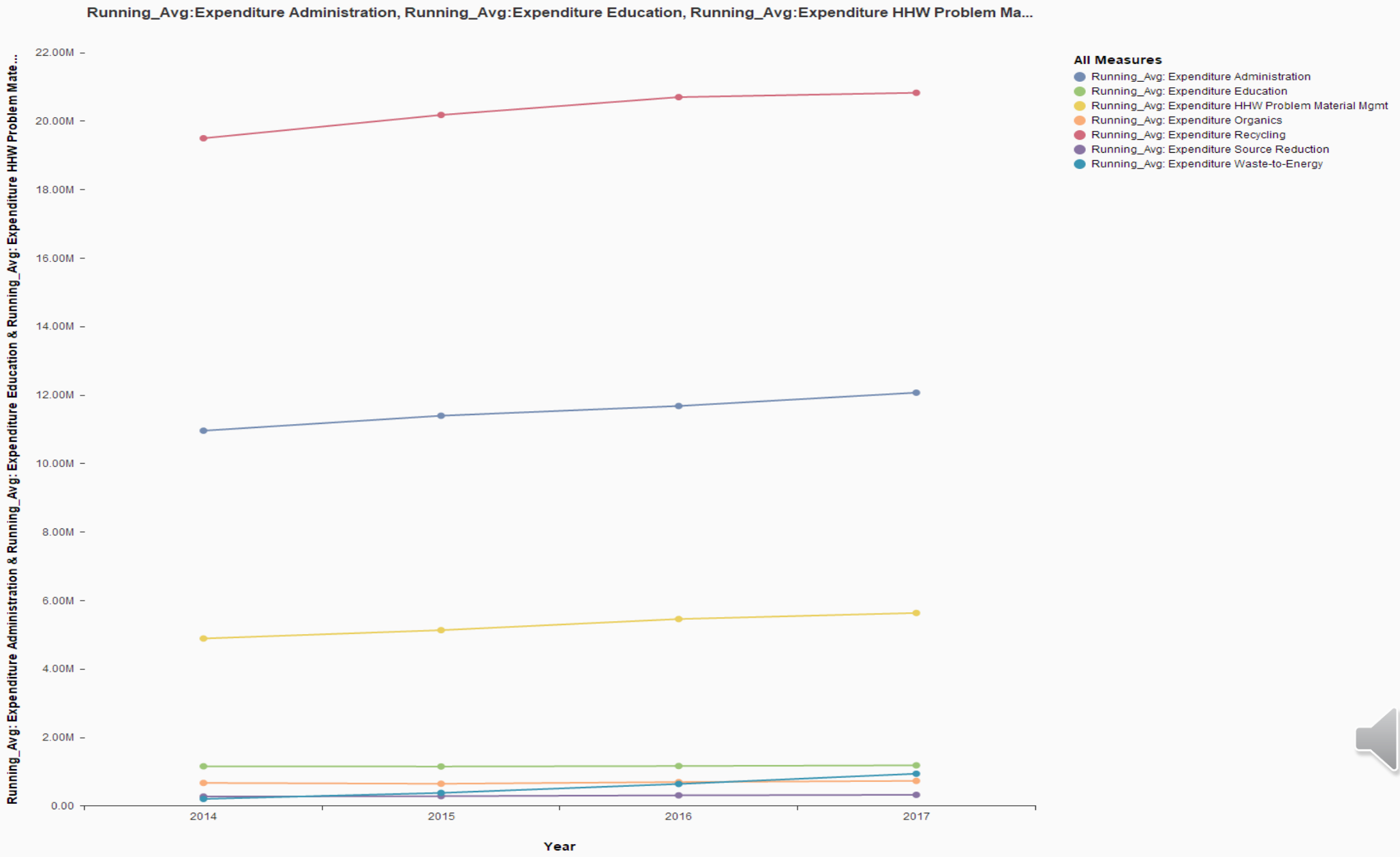
Question 3:

Are there different recycling patterns between counties for organics, WTE, landfilled, and other waste methods?

How:

- Line chart
- Measures & dimensions




Running_Avg:Expenditure Administration, Running_Avg:Expenditure Education, Running_Avg:Expenditure HHW Problem Ma...
Running_Avg: Expenditure Administration & Running_Avg: Expenditure Education & Running_Avg: Expenditure HHW Problem Mate...
22.00M
20.00M
18.00M
16.00M
14.00M
12.00M
10.00M
8.00M
6.00M
4.00M
2.00M
0.00
2014
2015
2016
2017
Year
All Measures
Running_Avg: Expenditure Administration
Running_Avg: Expenditure Education
Running_Avg: Expenditure HHW Problem Material Mgmt
Running_Avg: Expenditure Organics
Running_Avg: Expenditure Recycling
Running_Avg: Expenditure Source Reduction
Running_Avg: Expenditure Waste-to-Energy

AVERAGE EXPENDITURE: CATEGORY & YEAR

Answer:

- Statistical difference: $p<0.001$
- Differ by category and year
- Highest average spending: Recycling
- Second highest: Administration
- Fastest growing: WTE
- Consistently high: HHW Problem Material Mgmt

Averages (2017):

- Recycling: 268,467
- Administration: 167,589
- HHW Problem Material Mgmt: 78,096
- Waste-to-Energy: 23,281

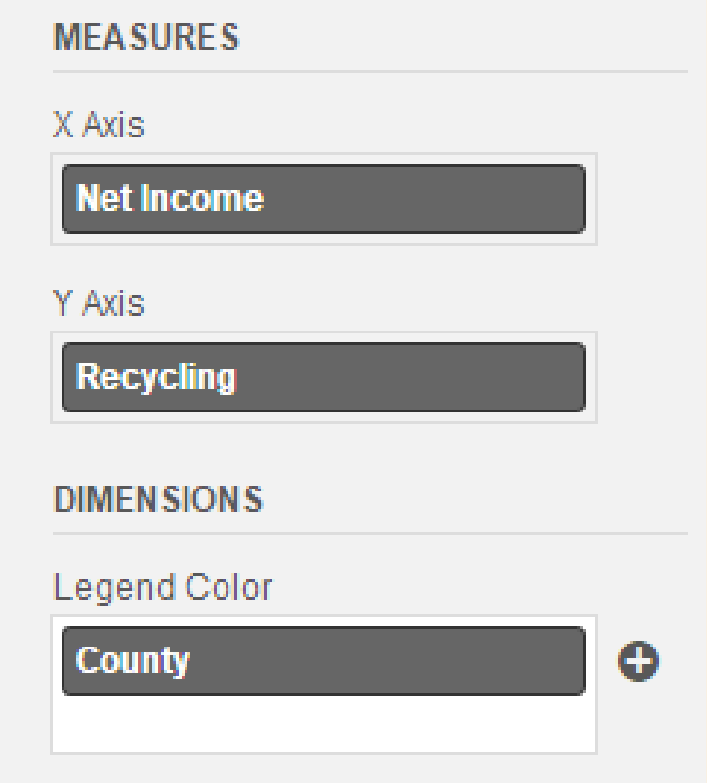
NET INCOME & RECYCLING

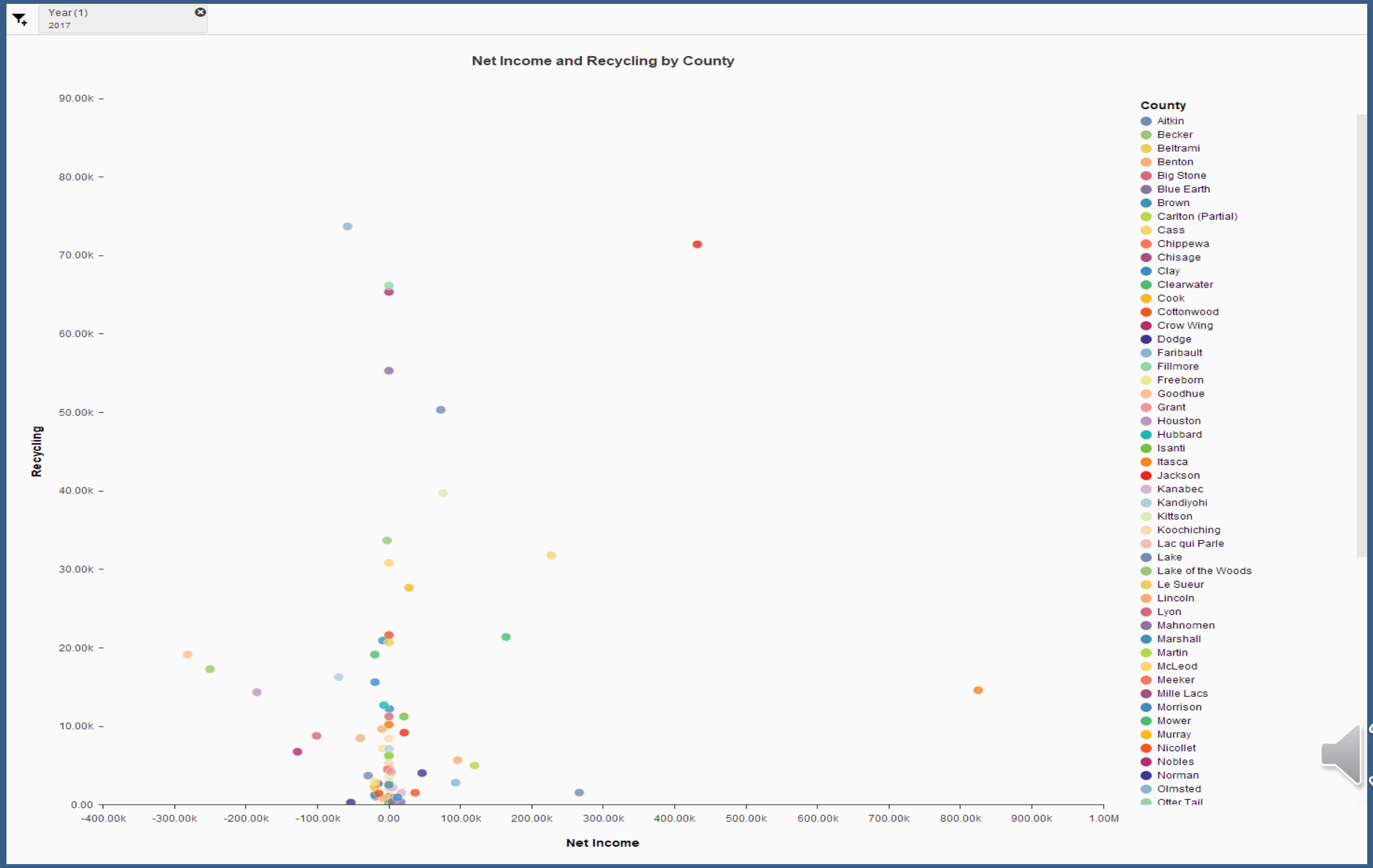
Question 4:

Does net income, revenue minus expenditure, impact recycling totals?

How:

- Scatterplot
- Measures & dimensions




Year (1)
2017
Net Income and Recycling by County
Recycling
90.00k
80.00k
70.00k
60.00k
50.00k
40.00k
30.00k
20.00k
10.00k
0.00
-400.00k
-300.00k
-200.00k
-100.00k
0.00
100.00k
200.00k
300.00k
400.00k
500.00k
600.00k
700.00k
800.00k
900.00k
1.00M
Net Income
County
Aitkin
Becker
Beltrami
Benton
Big Stone
Blue Earth
Brown
Carlton (Partial)
Cass
Chippewa
Chisage
Clay
Clearwater
Cook
Cottonwood
Crow Wing
Dodge
Faribault
Fillmore
Freeborn
Goodhue
Grant
Houston
Hubbard
Isanti
Itasca
Jackson
Kanabec
Kandiyohi
Kittson
Koochiching
Lac qui Parle
Lake
Lake of the Woods
Le Sueur
Lincoln
Lyon
Mahnomen
Marshall
Martin
McLeod
Meeker
Mille Lacs
Morrison
Mower
Murray
Nicollet
Nobles
Norman
Olmsted
Otter Tail

COUNTY PATTERNS: RECYCLING CATEGORY

Answer: No.

1. **Why:**
2. No clear upward trend
3. Not consistent between high income counties and recycling
4. Weak/unclear relationship

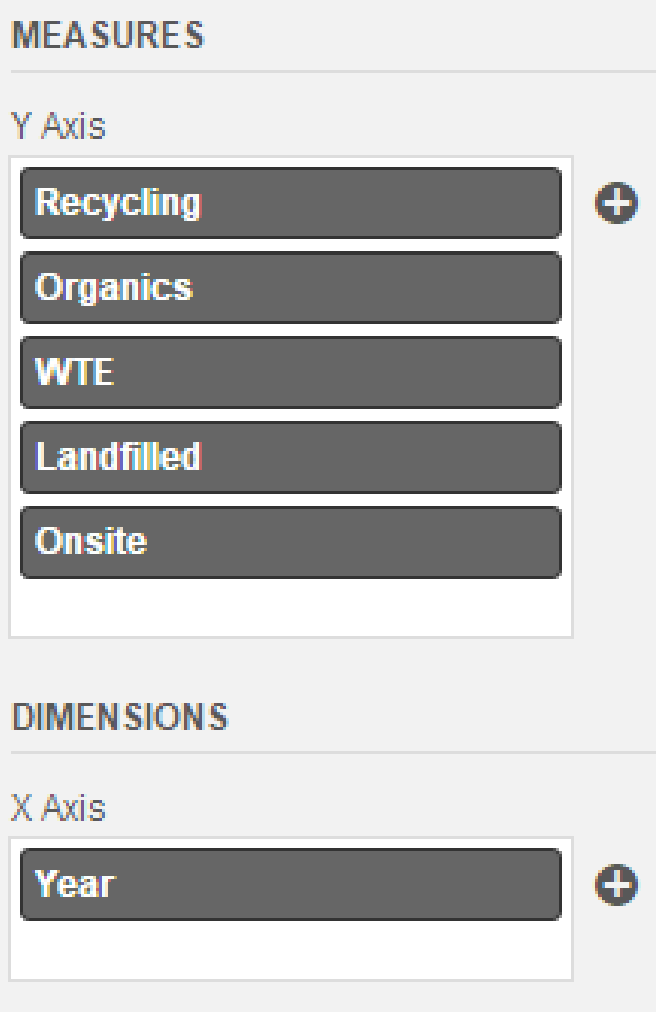
POLICY CHANGES & MINNESOTA TRENDS

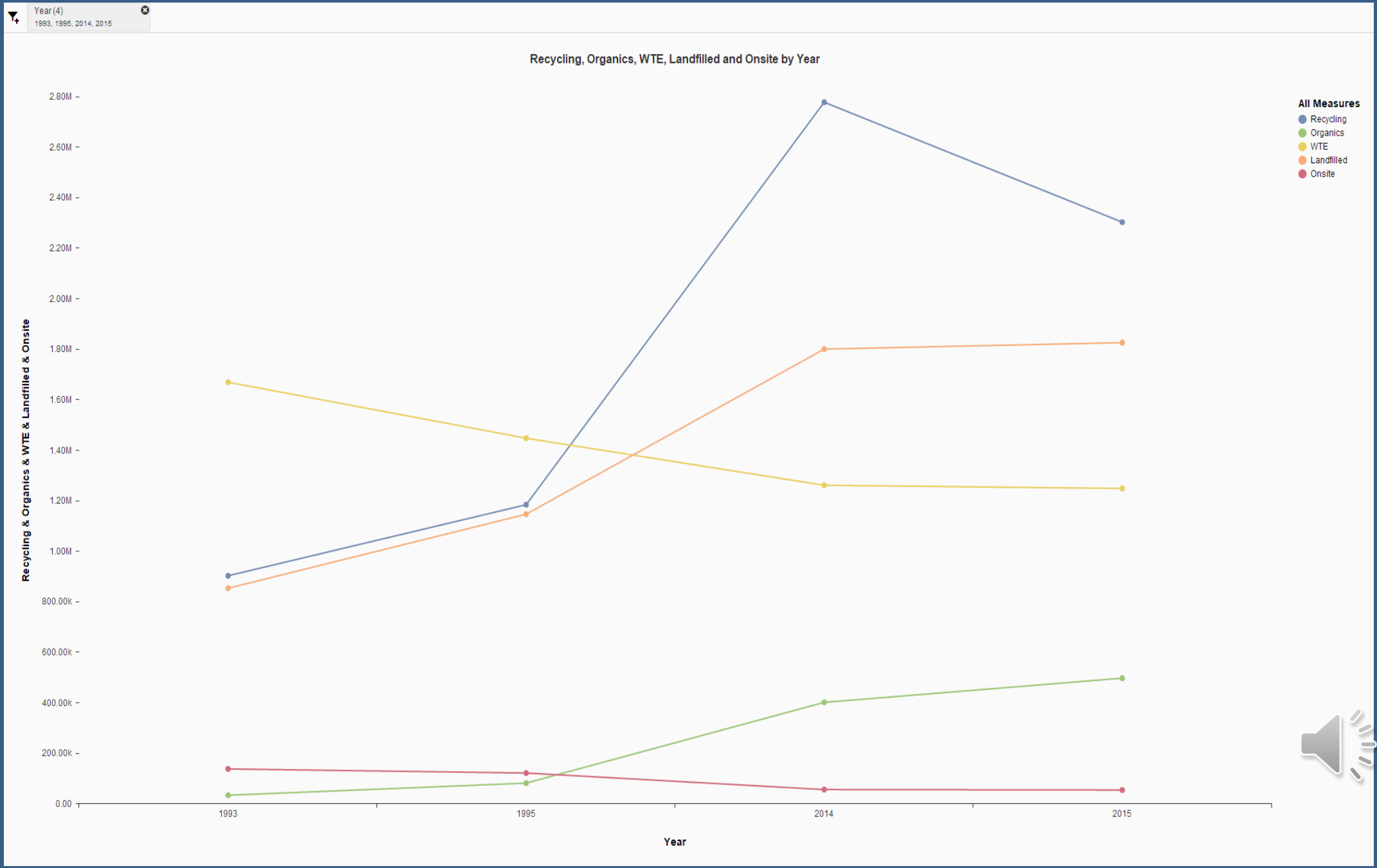
Question 5:

Examine patterns in the data related to policy changes in the history of recycling.

How:

- Line chart
- Measures & dimensions




Year (4)
1993, 1995, 2014, 2015
Recycling, Organics, WTE, Landfilled and Onsite by Year
Recycling & Organics & WTE & Landfilled & Onsite
2.80M
2.60M
2.40M
2.20M
2.00M
1.80M
1.60M
1.40M
1.20M
1.00M
800.00k
600.00k
400.00k
200.00k
0.00
1993
1995
2014
2015
Year
All Measures
Recycling
Organics
WTE
Landfilled
Onsite

POLICY CHANGES & MINNESOTA TRENDS

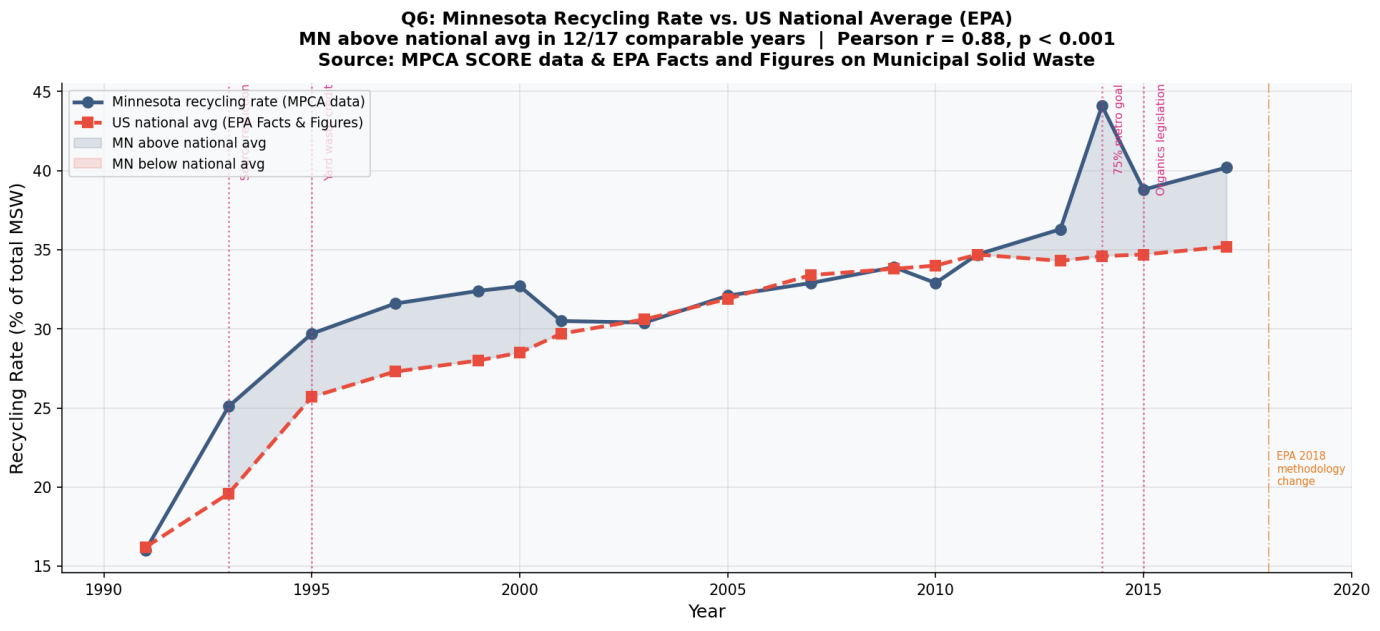
Answer:

- 1991: Waste hierarchy prioritizes composting over incineration and land disposal
- 1992: Yard waste disposal prohibited
- 1993: Source reduction credit added to goals
- 1995: Yard waste credit added and goals raised
- 2014: 75% metro / 35% greater MN goals, compostables counted, SCORE funding increased, broader recycling requirements
- 2015: Recyclable-materials definition expanded to include source-separated compostables and food waste streams

NATIONAL COMPARISON & MINNESOTA TRENDS

Question 6:

Do your own research on national recycling data and find the national pattern of recycling (please do not forget about explaining your dataset and source of it). Is that consistent with Minnesota's pattern?



* EPA changed food waste measurement methodology in 2018 — national rate dropped from 35.2% to 32.1%, reflecting broader food tracking, not a real recycling decline. 2017 used as final comparison year.

DATASET

EPA MSW Facts & Figures

SOURCE

epa.gov/facts-and-figures

U.S. RATE (2018)

Recycling + Composting

32.1%

MN RATE (2017)

Recycling + Composting

40.2%

Comparison

MN Above in 5/6 Yrs

NATIONAL COMPARISON & MINNESOTA TRENDS

Answer:

- Partly consistent — same upward trend nationally and in Minnesota
- MN above national average in 11 of 15 comparable years
- Gap widened significantly post-2014 due to state legislation
- Strong correlation confirmed: $r = 0.89$, $p < 0.001$
- EPA changed food waste methodology in 2018 — 2017 is the cleanest comparison year

MN vs. EPA recycling rate (%):

Year	Minnesota	US (EPA)	vs. National
1991	16.0%	16.2%	Below
1995	29.7%	25.7%	Above
2005	32.1%	31.9%	Above
2013	36.3%	34.3%	Above
2014	**44.1%**	**34.6%**	**Above**
2017	40.2%	35.2%	Above

Source: EPA Facts & Figures on Municipal Solid Waste — epa.gov/facts-and-figures-about-materials-waste-and-recycling

KEY FINDINGS

6 Questions · 3 Datasets · 27 Years of Data

Minnesota Recycling Patterns & Financial Trends

Q1 WASTE METHODS

Metro counties dominate all methods

Hennepin & Ramsey account for the largest waste volumes across all 5 methods. Patterns are only partially consistent — rural counties rely more heavily on landfilling.

Q2 RECYCLING CATEGORIES

Paper leads statewide; organics vary widely

Paper is the top recycled material in nearly every county. Organics recycling is highly concentrated in select metro counties, showing strong inconsistency across the state.

Q3 EXPENDITURE TRENDS

Recycling spending is highest; WTE grew fastest

Avg recycling expenditure ($268K) far exceeds all categories. Statistical difference confirmed at $p<0.001$. WTE showed the fastest growth from 2014–2017.

Q4 NET INCOME VS. RECYCLING

No significant correlation found

Scatter analysis shows a weak, unclear relationship. Higher net income does not predict higher recycling tonnage — funding structure alone doesn't drive output.

Q5 POLICY IMPACT

2014–2015 policies drove visible shifts

Recycling tonnage surged sharply around 2014, coinciding with the state's 75% recycling goal and the 2015 organics legislation separating organics from general recycling.

Q6 NATIONAL COMPARISON

Minnesota tracks above national EPA average

Using EPA municipal solid waste data, Minnesota's recycling rate exceeds the national average, particularly after 2014 policy changes accelerated statewide volume.

BY THE NUMBERS

The most important numbers from our analysis

86

counties

analyzed across all
3 MPCA datasets

27

years

of waste management
data (1991–2017)

$268K

avg/county

spent on recycling
(highest category, 2017)

p<0.001

significance

expenditure differences
statistically confirmed

Policy impact is real

Recycling tonnage increased sharply after the 2014 state goal and 2015 organics legislation — the most visible trend in the 27-year dataset.

Money ≠ more recycling

Net income (revenue minus expenditure) shows no significant correlation with total recycling tonnage across Minnesota counties.

Minnesota outperforms nationally

Compared to EPA national averages, Minnesota's recycling rate is above the national benchmark, especially post-2014.

LIMITATIONS

01 Financial data window

Expenditure & Revenue dataset covers only 2014–2017, restricting the net income analysis to 4 years. All other datasets span 1991–2017.

02 EPA methodology shift (2018)

The EPA changed its food waste measurement methodology in 2018, making 2017 the cleaner and more defensible comparison year for this project. Minnesota and EPA data are comparable but not perfectly identical in every category definition.

03 County name inconsistencies

County names varied slightly across datasets (e.g. "Carlton (Partial)" vs. "Carlton - partial"), requiring manual cleaning before datasets could be merged for Q4 analysis.

CONCLUSION

Minnesota Recycling Patterns & Financial Trends

All six MPCA questions addressed with appropriate visualizations and statistical methods — from county-level patterns to national benchmarking.

Local patterns fully mapped

The Minnesota analysis reveals distinct recycling behaviors across counties, material categories, expenditure trends, and policy timing — building a complete picture of statewide waste management.

Financial relationship clarified

Net income does not significantly predict recycling output. Spending is concentrated in recycling and administration, with WTE growing fastest — insights useful for future budget planning.

National context established

The EPA comparison provides a national benchmark. Minnesota tracks above the national average, particularly after 2014 policy changes — a strong indicator of effective state-level legislation.

THANK YOU

REFERENCES

Sec. 115A.931 MN Statutes. (2025). Mn.gov. https://www.revisor.mn.gov/statutes/cite/115a.931

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